

## CHARLTON, Australia

WMO: 948390

Lat: 36.2847S

Lon: 143.3342E

Elev: 132

StdP: 99.75

Time Zone: 10.00 (AUS)

Period: 04-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | 0.0        | 1.1 | -5.1                          | 2.5 | 25.5 | -3.0 | 3.0 | 22.8 | 9.0                   | 11.5 | 8.2 | 11.9 | 0.5                   | 190  | 0.491 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 16.1                   | 38.9            | 18.8 | 36.3 | 18.2 | 34.1 | 17.8 | 21.5                | 29.6 | 20.4 | 29.3 | 19.5 | 29.5 | 4.7                  | 330  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.5                            | 14.5 | 23.2 | 18.0 | 13.1 | 22.9 | 16.6 | 12.0 | 21.9 | 63.1          | 28.3 | 59.4 | 29.2 | 56.1 | 29.5 | 25.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|--|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |  |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |  |
| 8.4               | 7.5  | 6.8 | DB                         | -2.3 | 43.0               | 1.4 | 1.5                                                | -3.3 | 44.1       | -4.1 | 45.0       | -4.9 | 45.9       | -5.8 | 47.0 |  |
|                   |      |     | WB                         | -2.4 | 23.1               | 1.5 | 1.3                                                | -3.4 | 24.0       | -4.3 | 24.7       | -5.1 | 25.4       | -6.1 | 26.3 |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 15.7   | 23.7  | 22.7  | 20.2  | 16.1  | 12.1  | 9.3   | 8.8   | 9.5   | 11.7  | 15.2  | 18.7  | 21.0 |
|                                                                                  | DBStd       | 6.31   | 4.39  | 4.08  | 4.14  | 3.45  | 2.67  | 2.04  | 2.11  | 2.34  | 3.04  | 4.01  | 4.50  | 4.32 |
|                                                                                  | HDD10.0     | 147    | 0     | 0     | 0     | 1     | 8     | 36    | 49    | 38    | 13    | 2     | 0     | 0    |
|                                                                                  | HDD18.3     | 1541   | 4     | 7     | 26    | 85    | 193   | 271   | 296   | 274   | 202   | 116   | 48    | 18   |
|                                                                                  | CDD10.0     | 2234   | 424   | 357   | 317   | 183   | 73    | 16    | 12    | 22    | 62    | 164   | 262   | 342  |
|                                                                                  | CDD18.3     | 586    | 169   | 131   | 84    | 17    | 1     | 0     | 0     | 0     | 2     | 19    | 60    | 102  |
|                                                                                  | CDH23.3     | 8336   | 2458  | 1692  | 1037  | 218   | 3     | 0     | 0     | 2     | 36    | 373   | 949   | 1568 |
|                                                                                  | CDH26.7     | 4176   | 1388  | 860   | 467   | 57    | 0     | 0     | 0     | 0     | 7     | 140   | 453   | 803  |
| Wind                                                                             | WSAvg       | 3.3    | 3.8   | 3.7   | 3.3   | 2.8   | 2.8   | 2.6   | 3.0   | 3.0   | 3.2   | 3.5   | 3.7   | 3.8  |
| Precipitation                                                                    | PrecAvg     | 368    | 21    | 21    | 18    | 20    | 31    | 45    | 41    | 36    | 45    | 28    | 33    | 29   |
|                                                                                  | PrecMax     | 630    | 150   | 71    | 70    | 73    | 78    | 118   | 90    | 97    | 144   | 109   | 114   | 117  |
|                                                                                  | PrecMin     | 204    | 0     | 0     | 0     | 0     | 0     | 15    | 8     | 1     | 15    | 0     | 0     | 0    |
|                                                                                  | PrecStd     | 115    | 30    | 21    | 16    | 20    | 21    | 26    | 19    | 24    | 30    | 27    | 27    | 26   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 42.7  | 40.8  | 37.8  | 31.6  | 23.7  | 17.4  | 17.2  | 21.9  | 27.3  | 34.0  | 37.6  | 40.7 |
|                                                                                  |             | MCWB   | 19.8  | 20.1  | 18.3  | 16.7  | 14.2  | 12.1  | 11.2  | 12.5  | 13.6  | 15.7  | 17.5  | 18.4 |
|                                                                                  | 2%          | DB     | 39.8  | 36.6  | 34.5  | 28.0  | 20.9  | 15.9  | 15.4  | 18.6  | 23.7  | 30.5  | 34.3  | 37.1 |
|                                                                                  |             | MCWB   | 19.2  | 19.1  | 18.1  | 15.7  | 13.3  | 11.4  | 11.5  | 11.6  | 13.4  | 15.1  | 17.0  | 17.8 |
|                                                                                  | 5%          | DB     | 36.5  | 34.2  | 31.6  | 25.8  | 18.9  | 14.8  | 14.0  | 16.4  | 20.9  | 27.5  | 31.8  | 34.1 |
|                                                                                  |             | MCWB   | 18.5  | 18.5  | 17.6  | 14.8  | 13.0  | 11.0  | 10.6  | 11.1  | 12.8  | 14.3  | 16.6  | 17.1 |
|                                                                                  | 10%         | DB     | 33.8  | 31.9  | 29.0  | 23.6  | 17.4  | 13.8  | 13.0  | 14.7  | 18.5  | 24.7  | 29.0  | 31.3 |
|                                                                                  |             | MCWB   | 18.0  | 17.7  | 16.9  | 14.2  | 12.4  | 10.7  | 10.1  | 10.4  | 11.7  | 13.8  | 15.8  | 16.5 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 23.1  | 22.7  | 20.9  | 18.0  | 16.0  | 13.8  | 13.4  | 13.9  | 16.1  | 18.5  | 20.5  | 21.6 |
|                                                                                  |             | MCDB   | 30.7  | 30.6  | 27.5  | 24.3  | 18.6  | 14.7  | 15.2  | 18.3  | 21.7  | 25.9  | 28.5  | 29.0 |
|                                                                                  | 2%          | WB     | 21.6  | 21.4  | 19.8  | 17.1  | 14.7  | 12.6  | 12.1  | 12.6  | 14.5  | 16.9  | 19.2  | 20.0 |
|                                                                                  |             | MCDB   | 30.8  | 29.5  | 28.5  | 22.9  | 18.6  | 14.4  | 14.0  | 16.9  | 20.7  | 24.6  | 27.9  | 28.1 |
|                                                                                  | 5%          | WB     | 20.5  | 20.1  | 18.8  | 16.1  | 13.9  | 11.9  | 11.2  | 11.6  | 13.4  | 15.6  | 18.0  | 18.8 |
|                                                                                  |             | MCDB   | 30.7  | 28.8  | 27.0  | 22.9  | 17.8  | 13.8  | 13.2  | 15.4  | 19.3  | 24.0  | 27.6  | 29.6 |
|                                                                                  | 10%         | WB     | 19.4  | 19.1  | 17.9  | 15.1  | 13.0  | 11.2  | 10.5  | 10.8  | 12.3  | 14.5  | 16.9  | 17.7 |
|                                                                                  |             | MCDB   | 31.4  | 29.5  | 27.0  | 22.4  | 16.5  | 13.2  | 12.4  | 14.1  | 17.7  | 23.2  | 26.0  | 28.7 |
| Mean Daily<br>Temperature<br>Range                                               |             | MDBR   | 16.1  | 15.2  | 13.8  | 13.1  | 10.5  | 9.5   | 8.8   | 10.4  | 12.7  | 15.3  | 15.5  | 15.9 |
|                                                                                  | 5% DB       | MCDBR  | 19.9  | 18.2  | 17.0  | 16.3  | 12.4  | 10.8  | 9.9   | 13.4  | 16.2  | 19.8  | 19.3  | 19.7 |
|                                                                                  |             | MCWBR  | 6.1   | 5.8   | 5.7   | 6.8   | 6.6   | 6.9   | 6.3   | 7.5   | 7.7   | 7.7   | 6.2   | 6.3  |
|                                                                                  | 5% WB       | MCDBR  | 14.8  | 14.2  | 13.2  | 12.9  | 10.1  | 9.0   | 8.2   | 11.3  | 13.4  | 15.6  | 15.0  | 14.3 |
|                                                                                  |             | MCWBR  | 5.9   | 5.6   | 5.2   | 5.8   | 5.7   | 6.3   | 5.7   | 6.9   | 7.3   | 7.2   | 5.6   | 5.9  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.335  | 0.332 | 0.318 | 0.307 | 0.297 | 0.288 | 0.291 | 0.299 | 0.323 | 0.329 | 0.338 | 0.328 |      |
|                                                                                  | taud        | 2.503  | 2.523 | 2.552 | 2.562 | 2.572 | 2.594 | 2.580 | 2.539 | 2.457 | 2.453 | 2.464 | 2.514 |      |
|                                                                                  | Ebn at Noon | 999    | 980   | 957   | 909   | 861   | 843   | 860   | 904   | 934   | 971   | 990   | 1011  |      |
|                                                                                  | Edh at Noon | 113    | 106   | 96    | 83    | 72    | 65    | 70    | 83    | 103   | 113   | 117   | 113   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 7.81   | 6.97  | 5.60  | 4.03  | 2.74  | 2.16  | 2.32  | 3.27  | 4.55  | 6.04  | 6.99  | 7.78  |      |
|                                                                                  | RadStd      | 0.37   | 0.33  | 0.26  | 0.19  | 0.14  | 0.18  | 0.13  | 0.23  | 0.32  | 0.42  | 0.41  | 0.34  |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (23 neighbors) | +0.47 | N/A     | N/A    | +0.87   | N/A   | N/A   | N/A         | -88     | +151    | +81     |

Nomenclature: See separate page